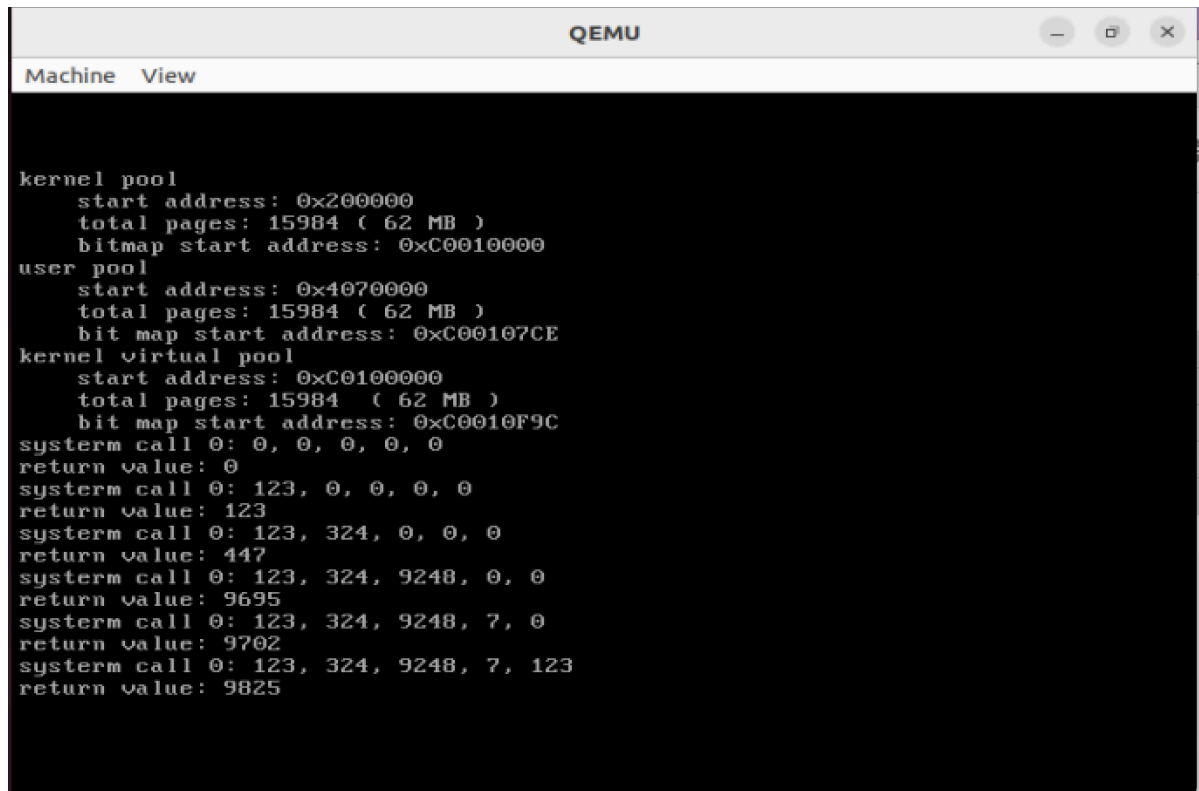


lab8

Assignment 1 — 系统调用与特权级转换

1.1 复现系统调用并分析特权级机制

复现 `src/2` 的系统调用代码：



```
QEMU
Machine View

kernel pool
  start address: 0x200000
  total pages: 15984 ( 62 MB )
  bitmap start address: 0xC0010000
user pool
  start address: 0x4070000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC00107CE
kernel virtual pool
  start address: 0xC0100000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC0010F9C
system call 0: 0, 0, 0, 0, 0
return value: 0
system call 0: 123, 0, 0, 0, 0
return value: 123
system call 0: 123, 324, 0, 0, 0
return value: 447
system call 0: 123, 324, 9248, 0, 0
return value: 9695
system call 0: 123, 324, 9248, 7, 0
return value: 9702
system call 0: 123, 324, 9248, 7, 123
return value: 9825
```

1. 编写至少 2 个自定义系统调用（不含已有的 `syscall_0`），例如：
 - 一个用于返回两个整数的最大值
 - 一个用于计算阶乘（迭代实现即可）
2. 在 `first_thread`（或进程）中调用你编写的系统调用，打印参数和返回值，截图展示正确性。

```

Booting from Hard Disk...
total memory: 133038080 bytes ( 126 MB )
kernel pool
  start address: 0x200000
  total pages: 15984 ( 62 MB )
  bitmap start address: 0xC0010000
user pool
  start address: 0x4070000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC00107CE
kernel virtual pool
  start address: 0xC0100000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC0010F9C
max(123, 456) = 456
factorial(5) = 120
max(999, 1) = 999
max(-5, -10) = -5
factorial(0) = 1
factorial(1) = 1
factorial(7) = 5040

```

3. 结合代码逐步分析系统调用的完整执行流程：

- 用户态： `asm_system_call` 如何将参数放入寄存器并执行 `int 0x80`
- 特权级转移：CPU 如何从 TSS 中获取内核栈地址，如何切换 SS/ESP
- 内核态： `asm_system_call_handler` 如何根据 `eax` 找到系统调用表中的函数并调用
- 返回： `iret` 如何恢复用户态的 CS/EIP/SS/ESP/EFLAGS

1.2 GDB 分析栈与段寄存器变化

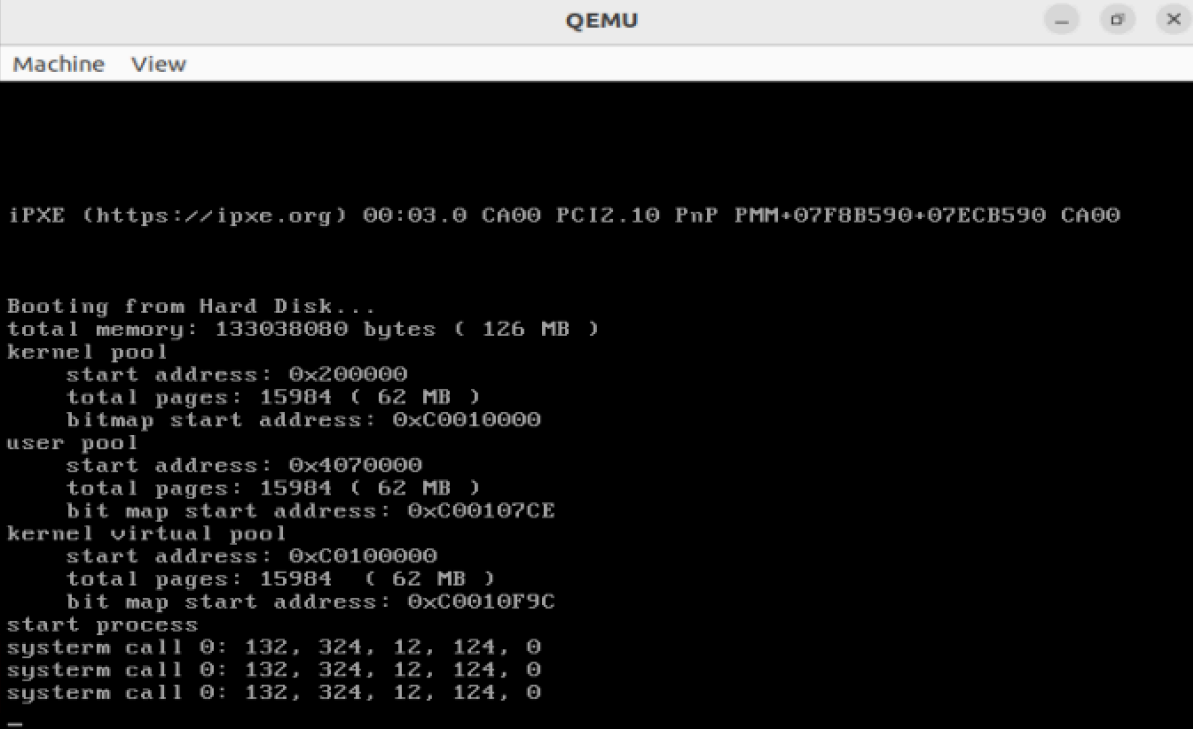
使用 GDB 调试 `src/3`（第一个进程），在 `int 0x80` 前后设置断点，回答以下问题：

1. 在执行 `int 0x80` 之前，CS、SS、ESP、EIP 的值分别是什么？CPL 是多少？
2. 在进入 `asm_system_call_handler` 之后，CS、SS、ESP 的值发生了什么变化？新的 SS 和 ESP 来自哪里？
3. 在执行 `iret` 之后，CS、SS、ESP 恢复为什么值？这些值保存在哪里？
4. 解释 TSS 中 `esp0` 和 `ss0` 字段在上述过程中的具体作用。

Assignment 2 — 进程创建与 Fork

2.1 进程的创建过程分析

复现 `src/3` 的代码

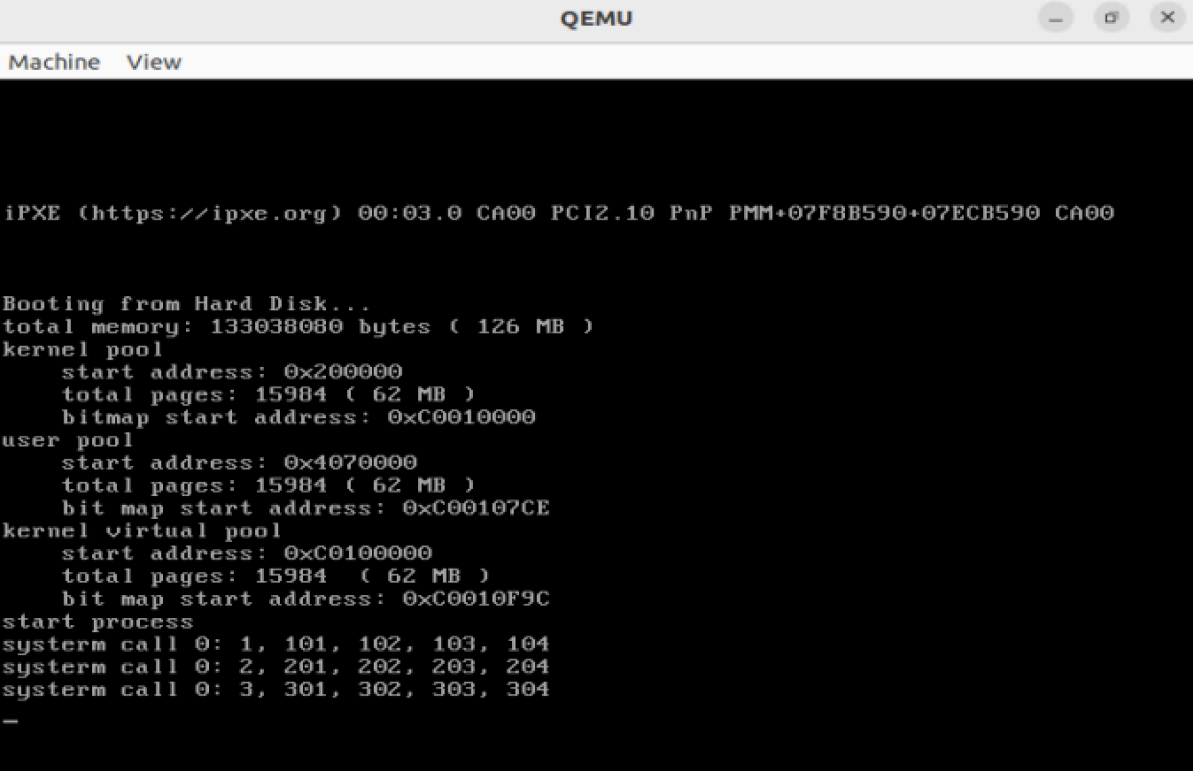
A screenshot of a QEMU window titled "QEMU" with standard window controls. The main area shows a black terminal with white text. The text displays the boot process of a system, starting with iPXE, then booting from a hard disk. It shows memory allocation for kernel and user pools, and the start of the process. The logs end with three "system call" lines, each with parameters 0, 132, 324, 12, 124, 0.

```
Machine  View

iPXE (https://ipxe.org) 00:03.0 CA00 PCI2.10 PnP PMM+07F8B590+07ECB590 CA00

Booting from Hard Disk...
total memory: 133038080 bytes ( 126 MB )
kernel pool
  start address: 0x2000000
  total pages: 15984 ( 62 MB )
  bitmap start address: 0xC0010000
user pool
  start address: 0x4070000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC00107CE
kernel virtual pool
  start address: 0xC0100000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC0010F9C
start process
system call 0: 132, 324, 12, 124, 0
system call 0: 132, 324, 12, 124, 0
system call 0: 132, 324, 12, 124, 0
—
```

在内核线程中创建多个用户进程（至少 3 个），完成以下内容：

A screenshot of a QEMU window titled "QEMU" with standard window controls. The main area shows a black terminal with white text. The text displays the boot process of a system, starting with iPXE, then booting from a hard disk. It shows memory allocation for kernel and user pools, and the start of the process. The logs end with three "system call" lines, each with parameters 0, 1, 101, 102, 103, 104; 0, 2, 201, 202, 203, 204; and 0, 3, 301, 302, 303, 304.

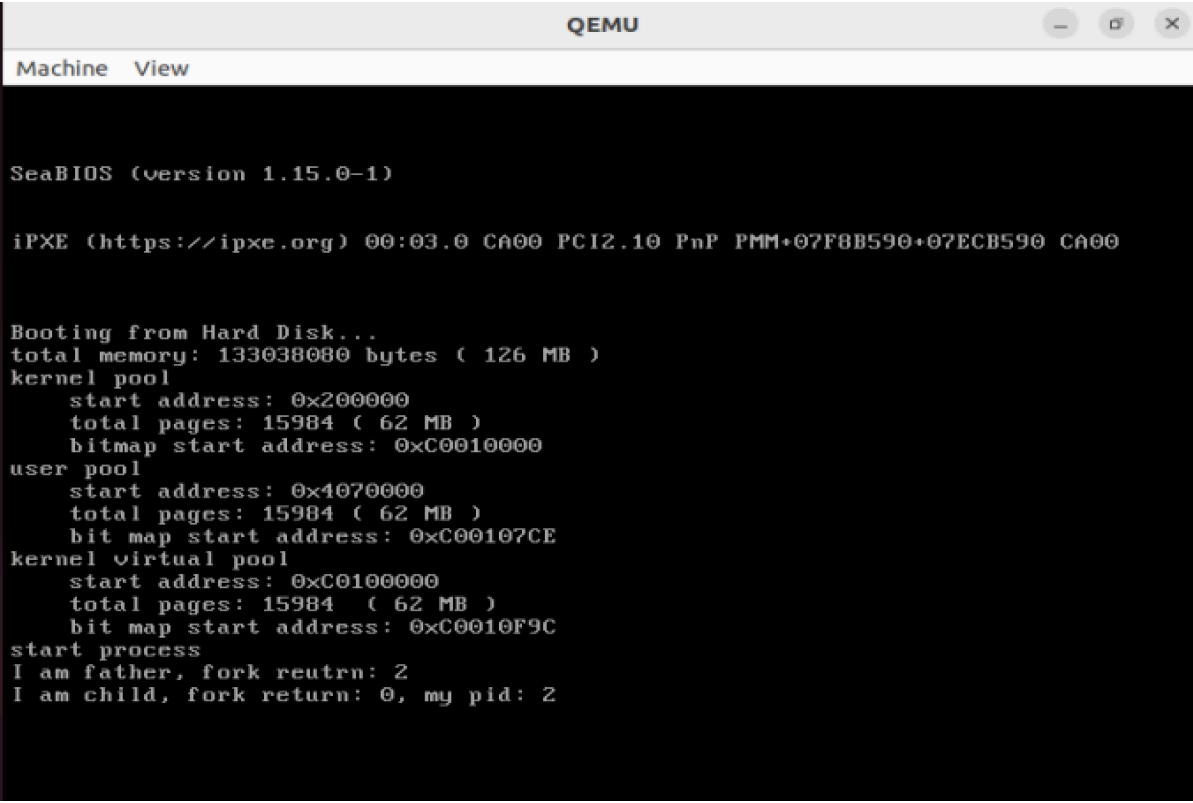
```
Machine  View

iPXE (https://ipxe.org) 00:03.0 CA00 PCI2.10 PnP PMM+07F8B590+07ECB590 CA00

Booting from Hard Disk...
total memory: 133038080 bytes ( 126 MB )
kernel pool
  start address: 0x2000000
  total pages: 15984 ( 62 MB )
  bitmap start address: 0xC0010000
user pool
  start address: 0x4070000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC00107CE
kernel virtual pool
  start address: 0xC0100000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC0010F9C
start process
system call 0: 1, 101, 102, 103, 104
system call 0: 2, 201, 202, 203, 204
system call 0: 3, 301, 302, 303, 304
—
```

2.2 Fork 的实现与分析

复现 `src/4` 的代码:

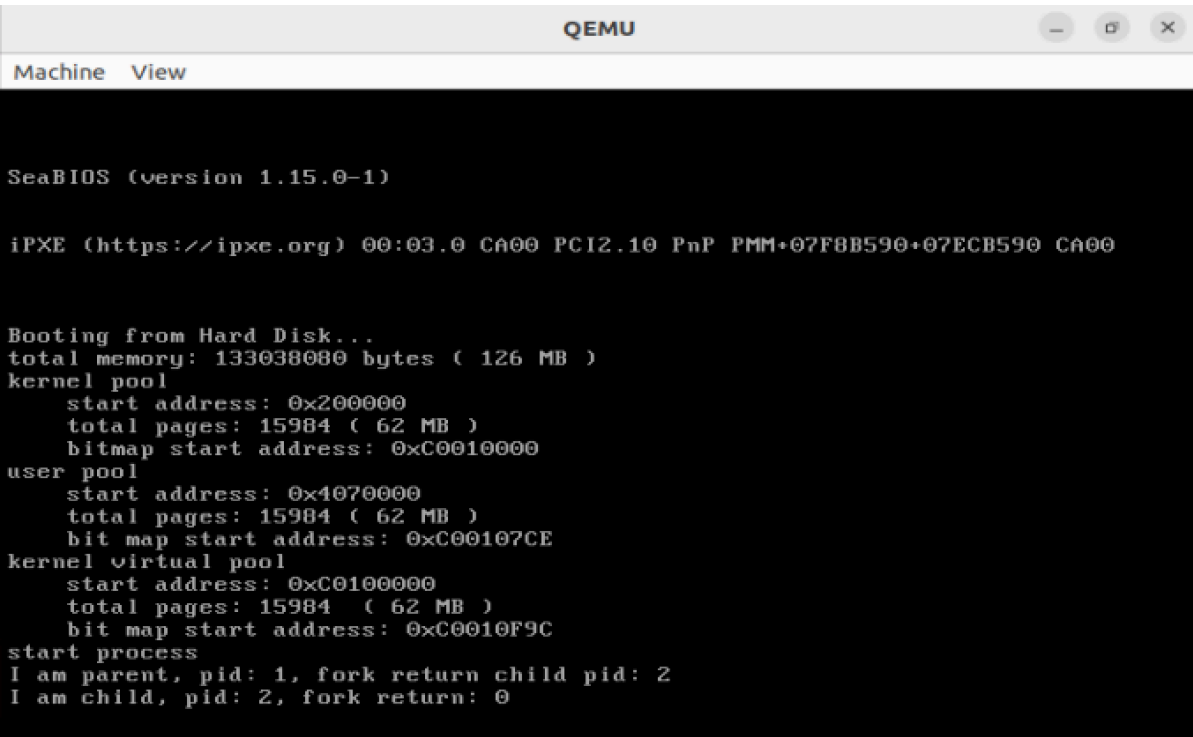


```
QEMU
Machine View

SeaBIOS (version 1.15.0-1)

iPXE (https://ipxe.org) 00:03.0 CA00 PCI2.10 PnP PMM+07F8B590+07ECB590 CA00

Booting from Hard Disk...
total memory: 133038080 bytes ( 126 MB )
kernel pool
  start address: 0x200000
  total pages: 15984 ( 62 MB )
  bitmap start address: 0xC0010000
user pool
  start address: 0x4070000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC00107CE
kernel virtual pool
  start address: 0xC0100000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC0010F9C
start process
I am father, fork reutrn: 2
I am child, fork return: 0, my pid: 2
```



```
QEMU
Machine View

SeaBIOS (version 1.15.0-1)

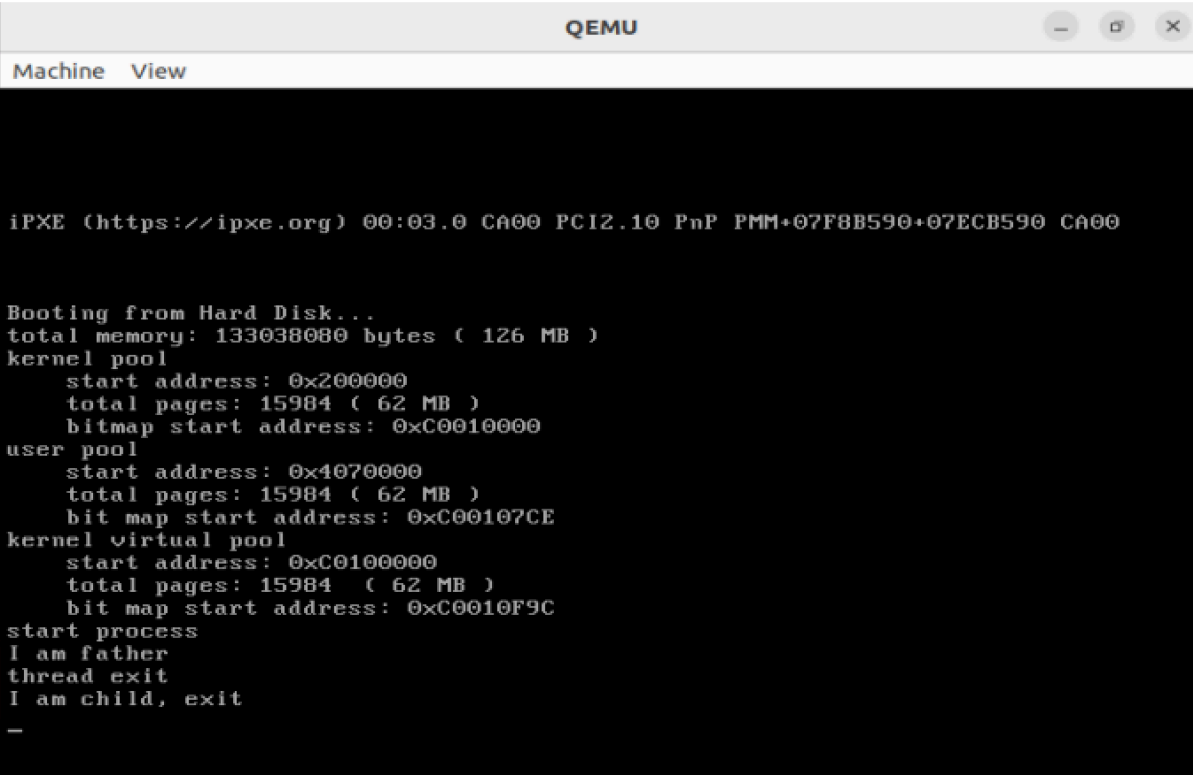
iPXE (https://ipxe.org) 00:03.0 CA00 PCI2.10 PnP PMM+07F8B590+07ECB590 CA00

Booting from Hard Disk...
total memory: 133038080 bytes ( 126 MB )
kernel pool
  start address: 0x200000
  total pages: 15984 ( 62 MB )
  bitmap start address: 0xC0010000
user pool
  start address: 0x4070000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC00107CE
kernel virtual pool
  start address: 0xC0100000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC0010F9C
start process
I am parent, pid: 1, fork return child pid: 2
I am child, pid: 2, fork return: 0
```

Assignment 3 — Wait/Exit 与进程生命周期

3.1 Exit 的实现与分析

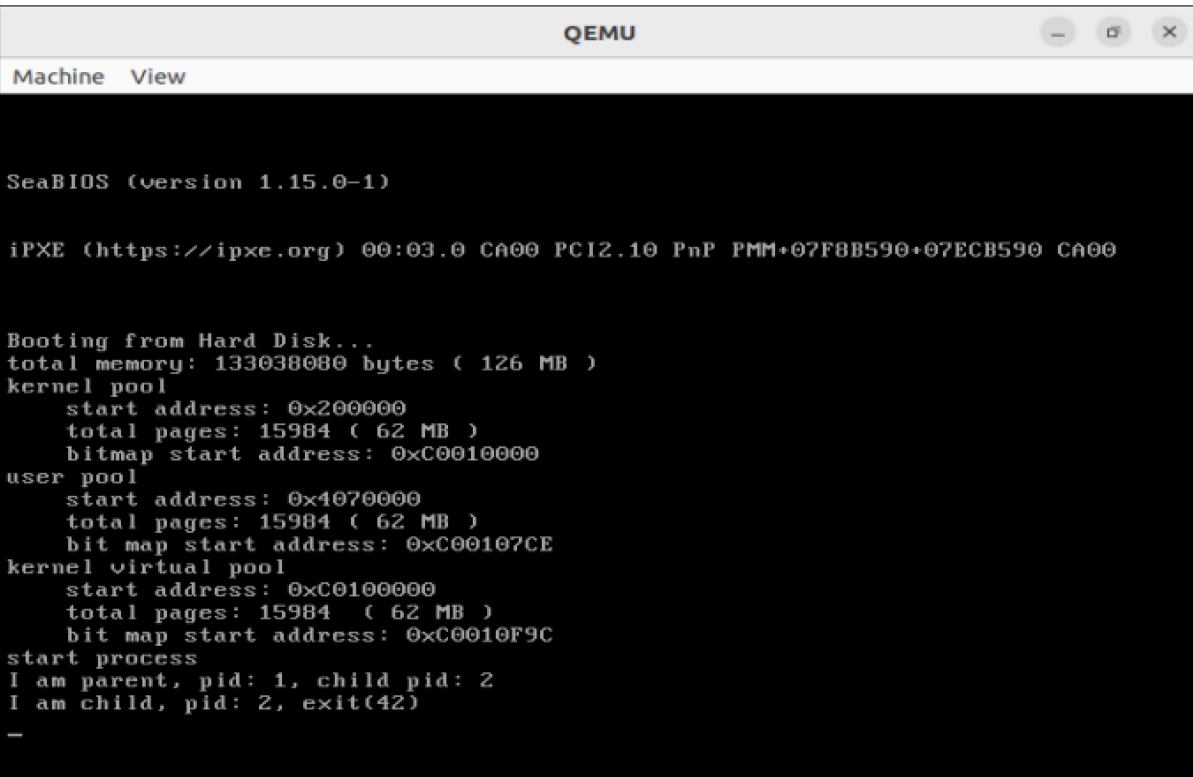
复现 `src/5` 的代码:



```
QEMU
Machine View

iPXE (https://ipxe.org) 00:03.0 CA00 PCI2.10 PnP PMM+07F8B590+07ECB590 CA00

Booting from Hard Disk...
total memory: 133038080 bytes ( 126 MB )
kernel pool
  start address: 0x2000000
  total pages: 15984 ( 62 MB )
  bitmap start address: 0xC0010000
user pool
  start address: 0x4070000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC00107CE
kernel virtual pool
  start address: 0xC0100000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC0010F9C
start process
I am father
thread exit
I am child, exit
-
```



```
QEMU
Machine View

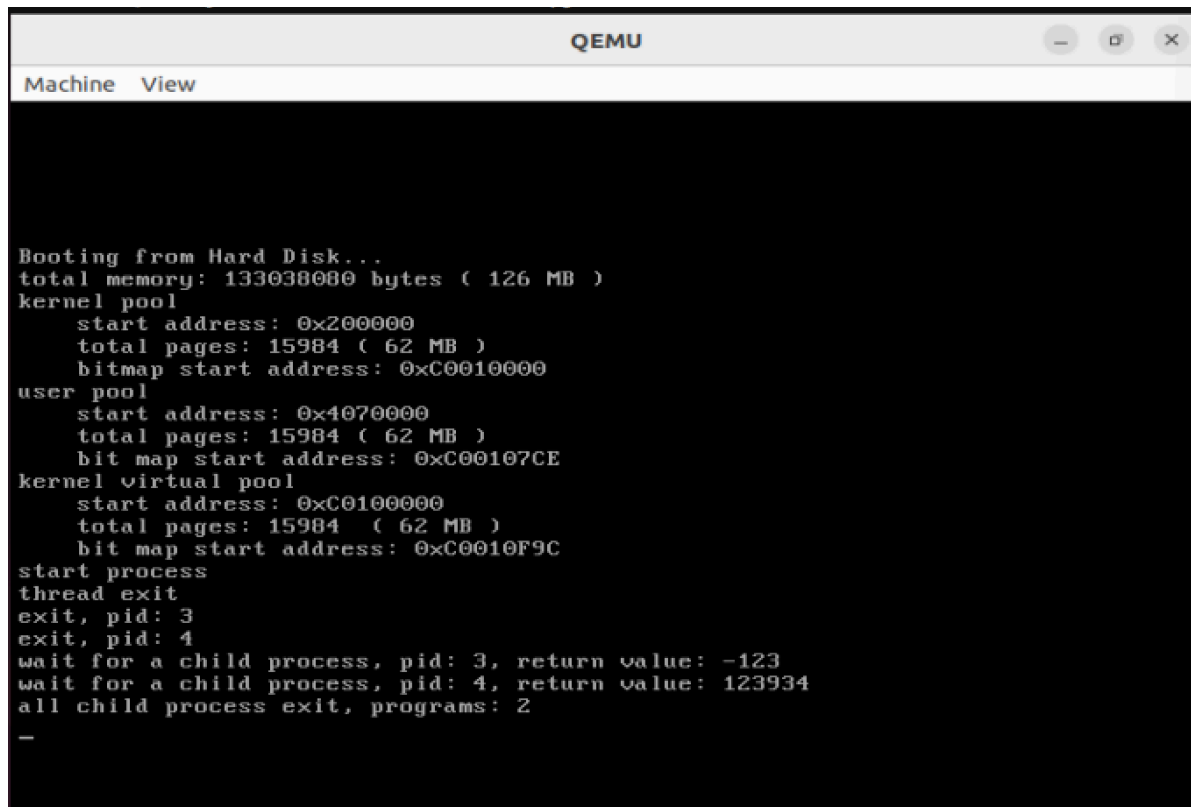
SeaBIOS (version 1.15.0-1)

iPXE (https://ipxe.org) 00:03.0 CA00 PCI2.10 PnP PMM+07F8B590+07ECB590 CA00

Booting from Hard Disk...
total memory: 133038080 bytes ( 126 MB )
kernel pool
  start address: 0x2000000
  total pages: 15984 ( 62 MB )
  bitmap start address: 0xC0010000
user pool
  start address: 0x4070000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC00107CE
kernel virtual pool
  start address: 0xC0100000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC0010F9C
start process
I am parent, pid: 1, child pid: 2
I am child, pid: 2, exit(42)
-
```

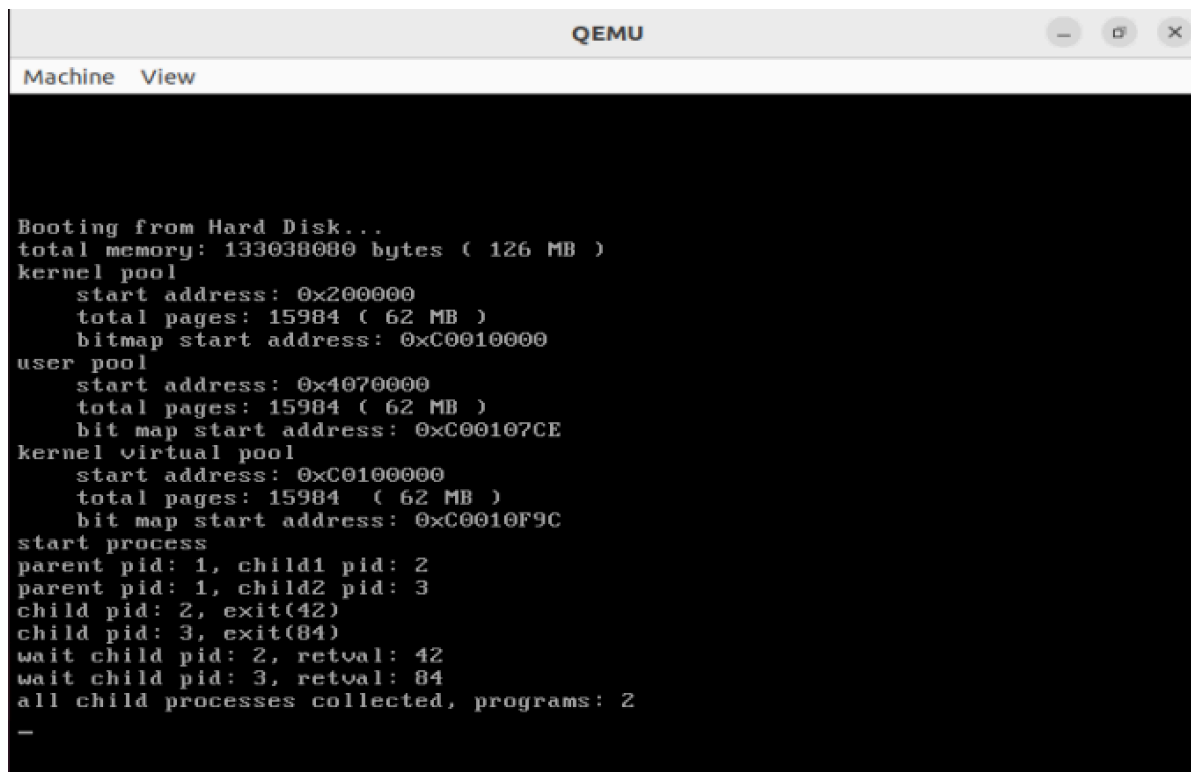
3.2 Wait 的实现与父子进程同步

复现 `src/6` 的代码:



```
QEMU
Machine View

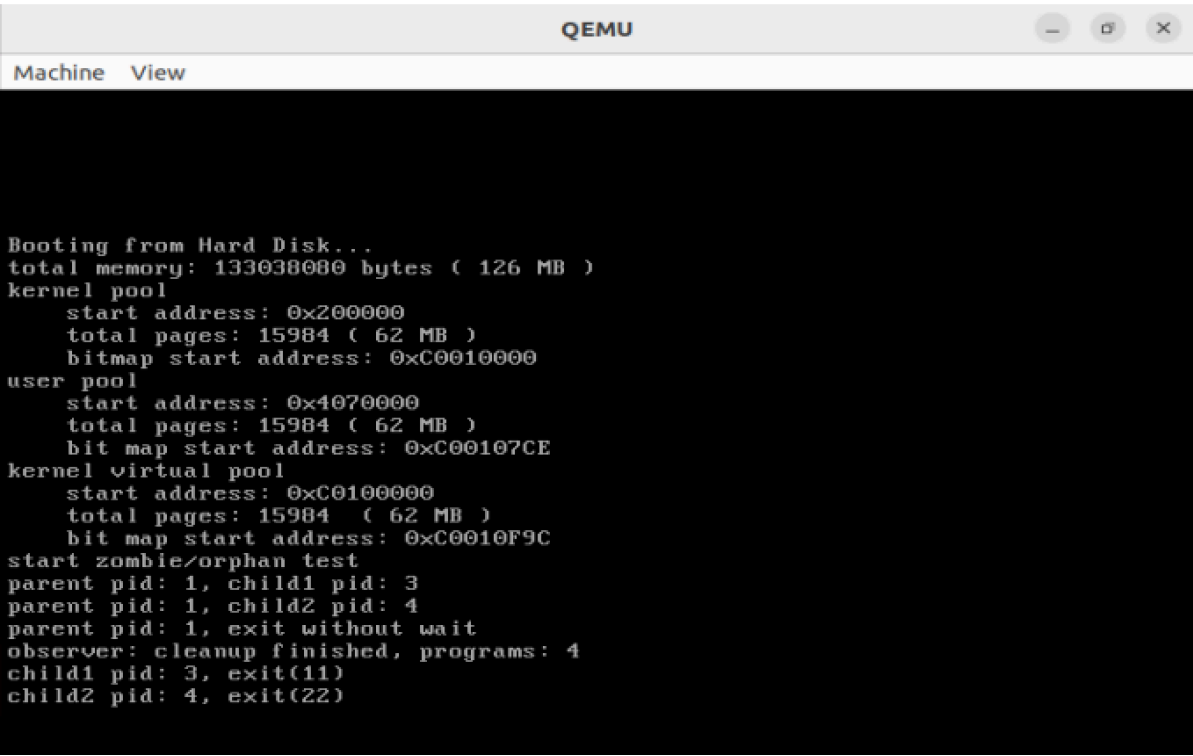
Booting from Hard Disk...
total memory: 133038080 bytes ( 126 MB )
kernel pool
  start address: 0x2000000
  total pages: 15984 ( 62 MB )
  bitmap start address: 0xC0010000
user pool
  start address: 0x4070000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC00107CE
kernel virtual pool
  start address: 0xC0100000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC0010F9C
start process
thread exit
exit, pid: 3
exit, pid: 4
wait for a child process, pid: 3, return value: -123
wait for a child process, pid: 4, return value: 123934
all child process exit, programs: 2
-
```



```
QEMU
Machine View

Booting from Hard Disk...
total memory: 133038080 bytes ( 126 MB )
kernel pool
  start address: 0x2000000
  total pages: 15984 ( 62 MB )
  bitmap start address: 0xC0010000
user pool
  start address: 0x4070000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC00107CE
kernel virtual pool
  start address: 0xC0100000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC0010F9C
start process
parent pid: 1, child1 pid: 2
parent pid: 1, child2 pid: 3
child pid: 2, exit(42)
child pid: 3, exit(84)
wait child pid: 2, retval: 42
wait child pid: 3, retval: 84
all child processes collected, programs: 2
-
```

3.3 僵尸进程与孤儿进程处理

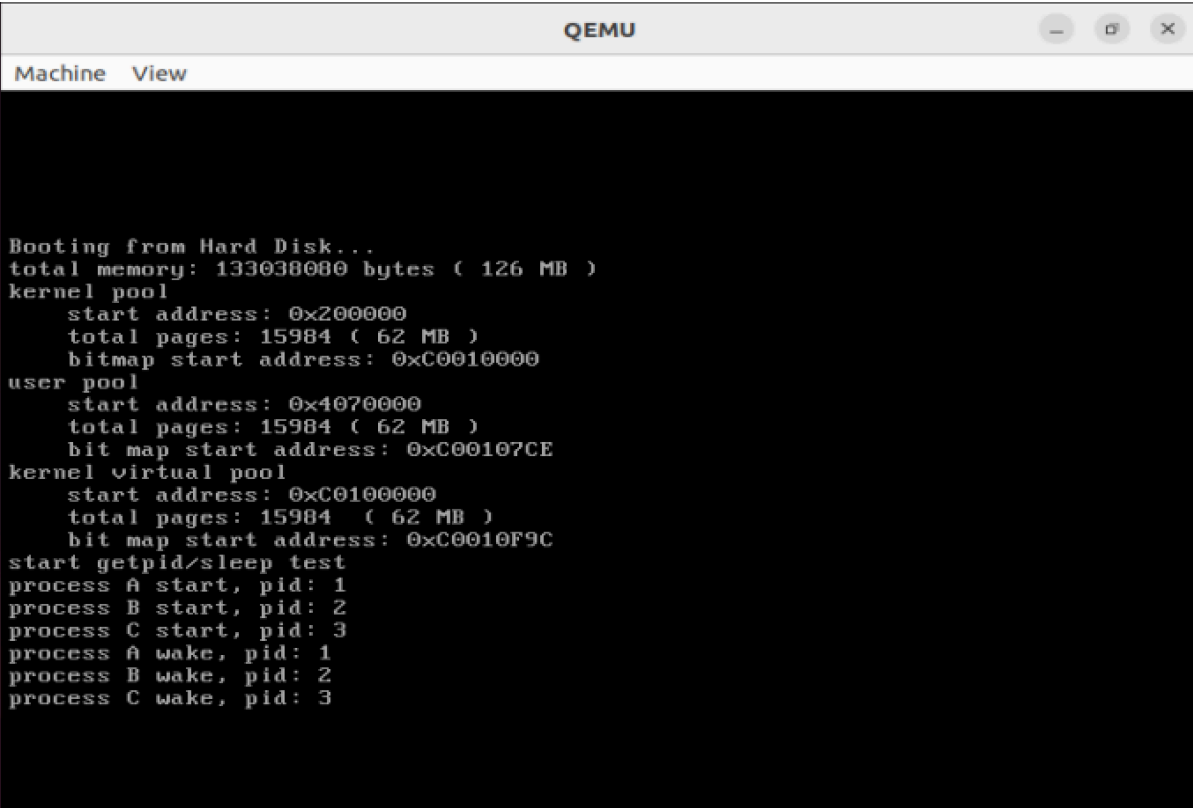


```
QEMU
Machine View

Booting from Hard Disk...
total memory: 133038080 bytes ( 126 MB )
kernel pool
  start address: 0x200000
  total pages: 15984 ( 62 MB )
  bitmap start address: 0xC0010000
user pool
  start address: 0x4070000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC00107CE
kernel virtual pool
  start address: 0xC0100000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC0010F9C
start zombie/orphan test
parent pid: 1, child1 pid: 3
parent pid: 1, child2 pid: 4
parent pid: 1, exit without wait
observer: cleanup finished, programs: 4
child1 pid: 3, exit(11)
child2 pid: 4, exit(22)
```

Assignment 4 — 选做

4.1 新增系统调用: getpid 和 sleep



```
QEMU
Machine View

Booting from Hard Disk...
total memory: 133038080 bytes ( 126 MB )
kernel pool
  start address: 0x200000
  total pages: 15984 ( 62 MB )
  bitmap start address: 0xC0010000
user pool
  start address: 0x4070000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC00107CE
kernel virtual pool
  start address: 0xC0100000
  total pages: 15984 ( 62 MB )
  bit map start address: 0xC0010F9C
start getpid/sleep test
process A start, pid: 1
process B start, pid: 2
process C start, pid: 3
process A wake, pid: 1
process B wake, pid: 2
process C wake, pid: 3
```